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UML Diagrams — MISSION OS

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1. Use Case Diagram

```
Parse error on line 2:
...ph TB      subgraph "MISSION OS Platform"
-----^
Expecting 'SEMI', 'NEWLINE', 'SPACE', 'EOF', 'GRAPH', 'DIR',
'TAGEND', 'TAGSTART', 'UP', 'DOWN', 'subgraph', 'end', 'SQE',
'PE', '-)', 'DIAMOND_STOP', 'MINUS', '--', 'ARROW_POINT',
'ARROW_CIRCLE', 'ARROW_CROSS', 'ARROW_OPEN',
'DOTTED_ARROW_POINT', 'DOTTED_ARROW_CIRCLE',
'DOTTED_ARROW_CROSS', 'DOTTED_ARROW_OPEN', '==',
'THICK_ARROW_POINT', 'THICK_ARROW_CIRCLE', 'THICK_ARROW_CROSS',
```

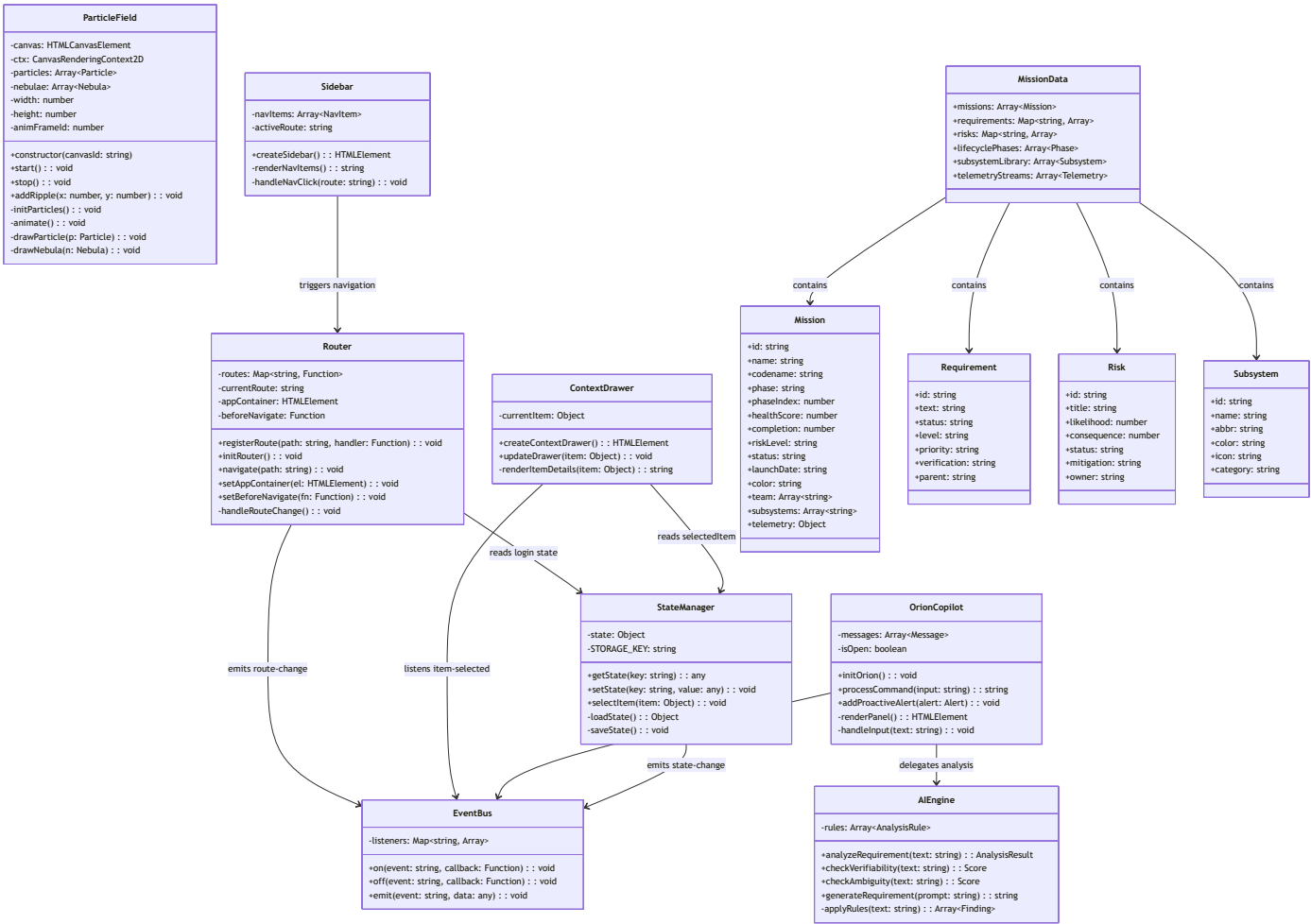
'THICK_ARROW_OPEN', 'PIPE', 'STYLE', 'LINKSTYLE', 'CLASSDEF',
'CLASS', 'CLICK', 'DEFAULT', 'NUM', 'PCT', 'COMMA', 'ALPHA',
'COLON', 'BRKT', 'DOT', 'PUNCTUATION', 'UNICODE_TEXT', 'PLUS',
'EQUALS', 'MULT', got 'STR'

Use Case Descriptions

ID	Use Case	Primary Actor	Description
UC-01	Login to System	All Users	Authenticate via codename and access code
UC-02	View Mission Dashboard	Mission Director	Monitor telemetry, stats, mission health
UC-03	Manage Requirements	Systems Engineer	View, analyze, and trace requirements
UC-04	Analyze Requirement Quality	Orion AI	AI-driven verifiability and ambiguity scoring
UC-05	Model System Architecture	Systems Engineer	Drag, zoom, pan subsystem topology
UC-06	View Engineering Diagrams	Systems Engineer	Browse 6 diagram types (WBS, PERT, etc.)
UC-07	Map Subsystem Interfaces	Systems Engineer	Interact with N ² matrix cells
UC-08	Manage Risks	Risk Manager	View heatmap, read risk register
UC-09	Conduct Trade Study	Systems Engineer	Adjust weights, compare alternatives
UC-10	Track V-Model Lifecycle	Test Engineer	Monitor phase progress, run gate checks
UC-11	Edit Documents	Systems Engineer	Select and edit mission documents
UC-12	Monitor Launch Countdown	Mission Director	Track countdown, simulate readiness

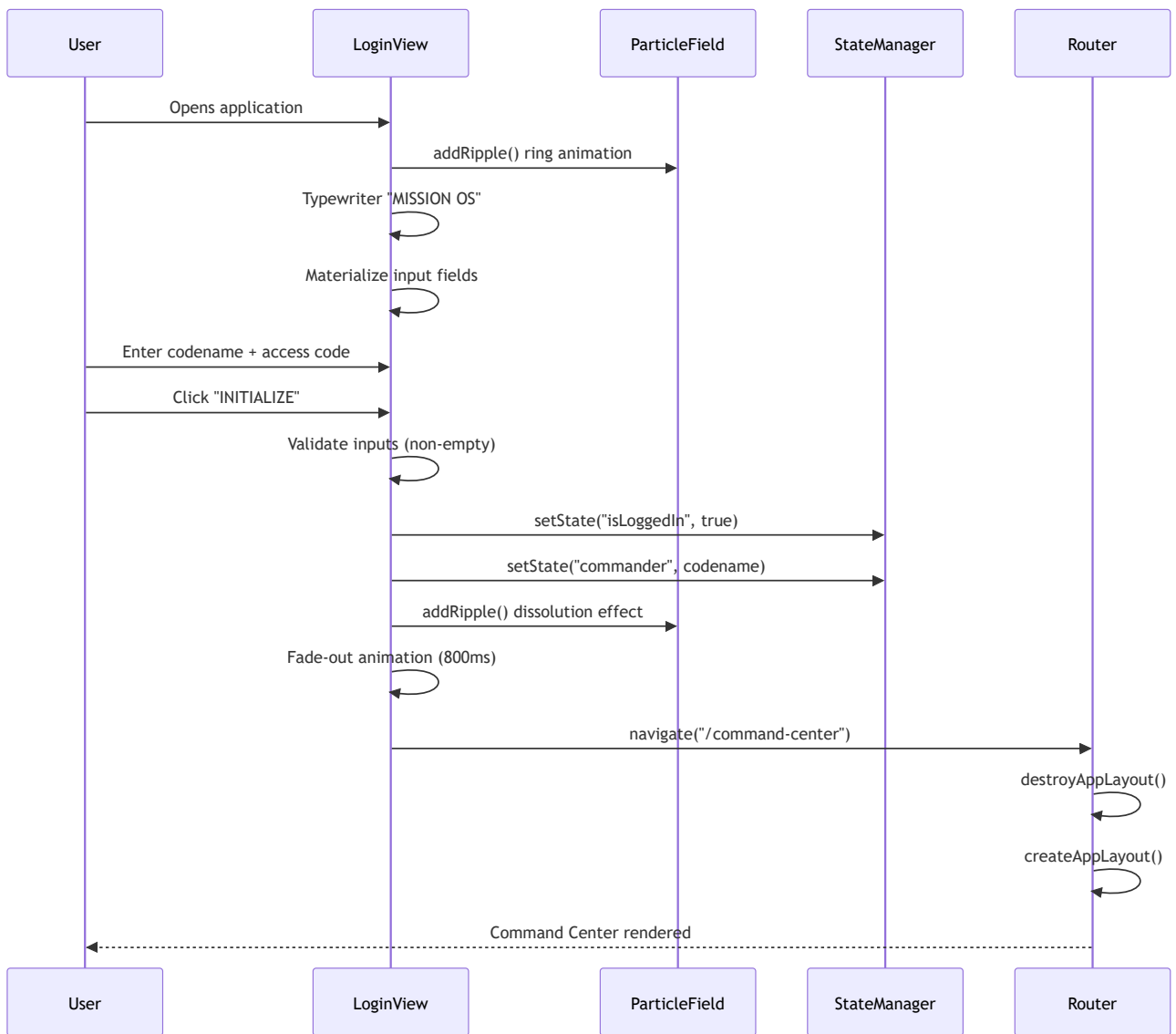
ID	Use Case	Primary Actor	Description
UC-13	Interact with AI Copilot	Systems Engineer	Issue commands, receive analysis
UC-14	View Item Context Details	All Users	Inspect selected item in context drawer
UC-15	Navigate Between Modules	All Users	Use sidebar to switch views

2. Class Diagram

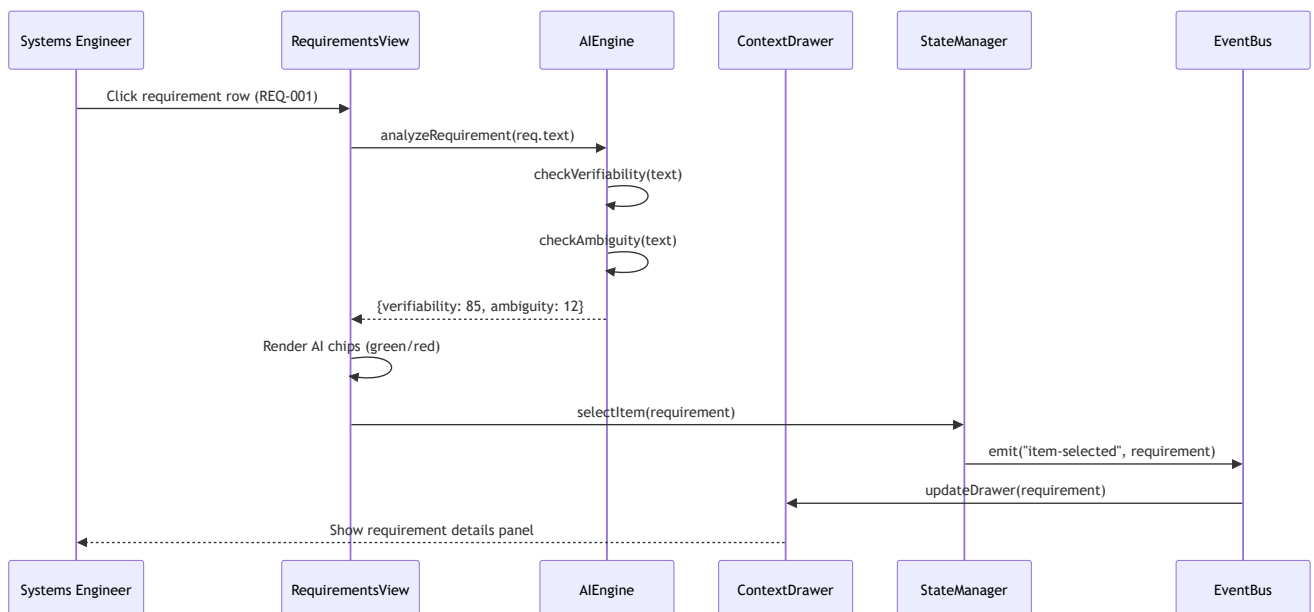


3. Sequence Diagrams

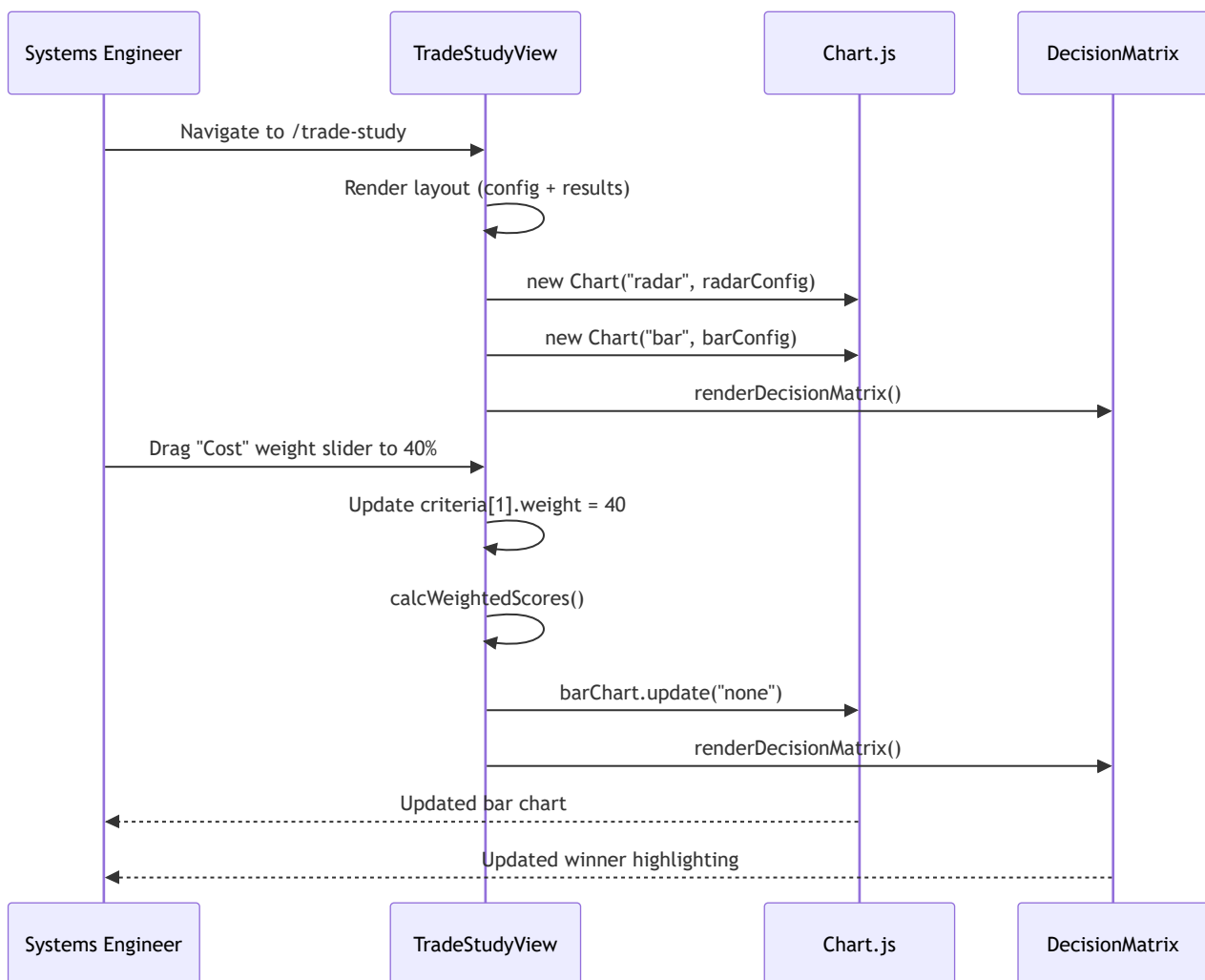
3.1 Login Sequence



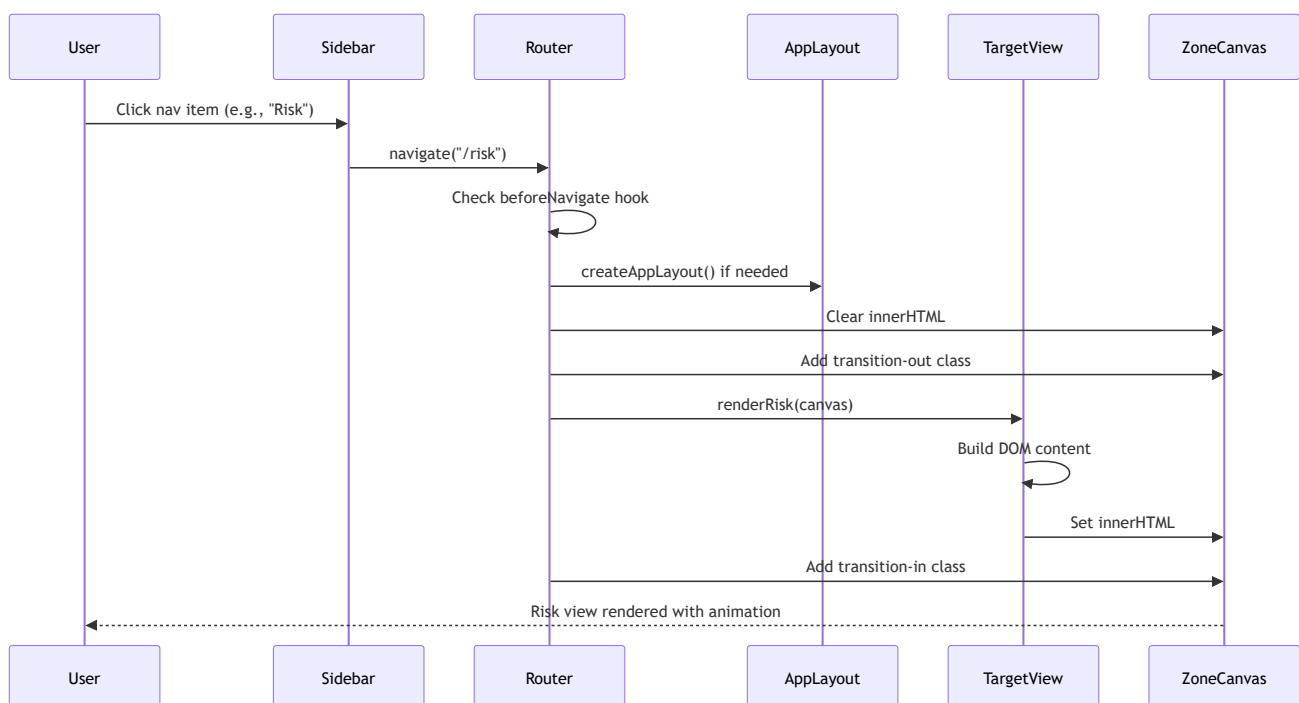
3.2 Requirements Analysis Sequence



3.3 Trade Study Interaction Sequence



3.4 Navigation Sequence



4. Activity Diagrams

4.1 Application Startup

```
Parse error on line 1:
flowchart TD      A([
^
Expecting 'NEWLINE', 'SPACE', 'GRAPH', got 'ALPHA'
```

4.2 Requirement Quality Analysis

```
Parse error on line 1:
flowchart TD      A([
^
Expecting 'NEWLINE', 'SPACE', 'GRAPH', got 'ALPHA'
```

5. Component Diagram

```
Parse error on line 2:
...ph TB      subgraph "Presentation Layer"
-----^
Expecting 'SEMI', 'NEWLINE', 'SPACE', 'EOF', 'GRAPH', 'DIR',
'TAGEND', 'TAGSTART', 'UP', 'DOWN', 'subgraph', 'end', 'SQE',
'PE', '-)', 'DIAMOND_STOP', 'MINUS', '--', 'ARROW_POINT',
'ARROW_CIRCLE', 'ARROW_CROSS', 'ARROW_OPEN',
'DOTTED_ARROW_POINT', 'DOTTED_ARROW_CIRCLE',
'DOTTED_ARROW_CROSS', 'DOTTED_ARROW_OPEN', '==',
'THICK_ARROW_POINT', 'THICK_ARROW_CIRCLE', 'THICK_ARROW_CROSS',
'THICK_ARROW_OPEN', 'PIPE', 'STYLE', 'LINKSTYLE', 'CLASSDEF',
'CLASS', 'CLICK', 'DEFAULT', 'NUM', 'PCT', 'COMMA', 'ALPHA',
'COLON', 'BRKT', 'DOT', 'PUNCTUATION', 'UNICODE_TEXT', 'PLUS',
'EQUALS', 'MULT', got 'STR'
```

6. State Diagram — Application State

Machine

Parse error on line 1:

```
stateDiagram-v2
```

```
^
```

Expecting 'NEWLINE', 'SPACE', 'GRAPH', got 'ALPHA'

7. Deployment Diagram

Parse error on line 2:

```
...ph TB      subgraph "Developer Machine"
```

```
-----^
```

Expecting 'SEMI', 'NEWLINE', 'SPACE', 'EOF', 'GRAPH', 'DIR',
'TAGEND', 'TAGSTART', 'UP', 'DOWN', 'subgraph', 'end', 'SQE',
'PE', '-)', 'DIAMOND_STOP', 'MINUS', '--', 'ARROW_POINT',
'ARROW_CIRCLE', 'ARROW_CROSS', 'ARROW_OPEN',
'DOTTED_ARROW_POINT', 'DOTTED_ARROW_CIRCLE',
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'THICK_ARROW_POINT', 'THICK_ARROW_CIRCLE', 'THICK_ARROW_CROSS',
'THICK_ARROW_OPEN', 'PIPE', 'STYLE', 'LINKSTYLE', 'CLASSDEF',
'CLASS', 'CLICK', 'DEFAULT', 'NUM', 'PCT', 'COMMA', 'ALPHA',
'COLON', 'BRKT', 'DOT', 'PUNCTUATION', 'UNICODE_TEXT', 'PLUS',
'EQUALS', 'MULT', got 'STR'